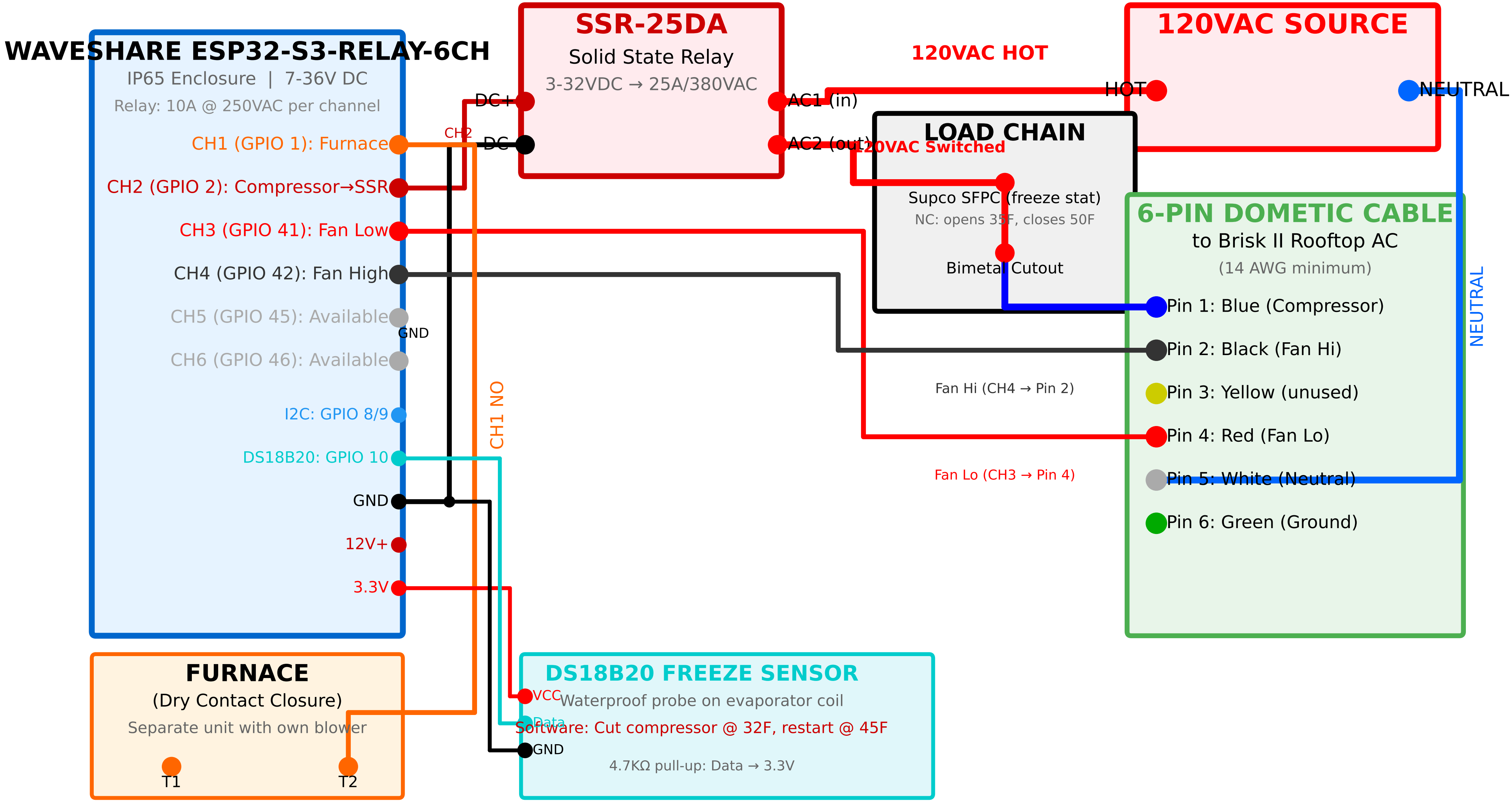
WAVESHARE ESP32-S3 — DIRECT 120VAC WIRING TO DOMETIC BRISK II

SSR-25DA for Compressor | Relay Channels for Fans + Furnace | Dual Freeze Protection



SAFETY CRITICAL — 120VAC WIRING

CH2 relay NO switches 12VDC milliamp trigger to SSR DC+
SSR-25DA switches 120VAC to compressor (heatsink required ~15W @ 12A)
CH3/CH4 switch 120VAC directly to fans (2-3A, 10A relay rating)
CH1 = dry contact for furnace (separate unit)
Dual freeze protection: DS18B20 software (32F) + Supco SFPC hardware (35F)
All 120VAC wiring: 14 AWG minimum, proper junction box per NEC

COMPONENT SPECIFICATIONS

WAVESHARE ESP32-S3-RELAY-6CH:
Power: 7-36V DC (or 5V USB-C) | IP65 (6.3" x 4.33" x 3.54")
Relay: 10A @ 250VAC/30VDC per channel (6 channels)
GPIO ASSIGNMENTS:
CH1=GPIO1 CH2=GPIO2 CH3=GPIO41 CH4=GPIO42 CH5=GPIO45 CH6=GPIO46
I2C: SDA=GPIO8, SCL=GPIO9 | 1-Wire: GPIO10 | Buzzer: GPIO21 | LED: GPIO38